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Nonparametric Predictive Inference for Discrete Data via Metropolis-Adjusted Dirichlet Sequences

Davide Agnoletto^a , Tommaso Rigon^b , and David B. Dunson^a

^aDepartment of Statistical Science, Duke University, Durham, NC; ^bDepartment of Economics, Management and Statistics, University of Milano–Bicocca, Milano, Italy

ABSTRACT

This article is motivated by challenges in conducting Bayesian inferences on unknown discrete distributions, with a particular focus on count data. To avoid the computational disadvantages of traditional mixture models, we develop a novel Bayesian predictive approach. In particular, our Metropolis-adjusted Dirichlet (MAD) sequence model characterizes the predictive measure as a mixture of a base measure and Metropolis-Hastings kernels centered on previous data points. The resulting MAD sequence is asymptotically exchangeable and the posterior on the data generator takes the form of a martingale posterior. This structure leads to straightforward algorithms for inference on count distributions, with easy extensions to multivariate, regression, and binary data cases. We obtain a useful asymptotic Gaussian approximation and illustrate the methodology on a variety of applications. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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1. Introduction

Bayesian nonparametric modeling of count distributions is a challenging task, and several solutions have been proposed over the years. A Bayesian nonparametric mixture of Poisson kernels lacks flexibility since both centering and scaling of the kernels are controlled by a single parameter. Canale and Dunson (2011), Canale and Prünster (2017) proposed a more flexible class of kernels obtained by rounding continuous distributions. However, estimating these mixture models using Markov chain Monte Carlo (MCMC) algorithms can be cumbersome in practice. Alternatively, one can directly specify a Dirichlet process (DP) prior (Ferguson 1973) on the data generator as

$$Y_i | P \stackrel{\text{iid}}{\sim} P, \quad P \sim \text{DP}(\alpha, P_0), \quad (1)$$

where $Y_i \in \mathcal{Y}$, $i = 1, \dots, n$, are assumed to be iid conditionally on P , $\mathcal{Y}$ is a countable space, α is the precision parameter, and P_0 is a base parametric distribution, such as a Poisson. Despite the appealing flexibility, it is well known that assuming (1) produces the conjugate posterior distribution

$$P | y_{1:n} \sim \text{DP}\left(\alpha + n, P_0 + \sum_{i=1}^n \delta_{y_i}\right),$$

with $y_{1:n} = (y_1, \dots, y_n)$, which lacks smoothing, in the sense that nearby points in the support are not encouraged to have similar probabilities. Such borrowing of information arises when there is some dependence, that is, positive correlation, between probabilities of nearby support points. However, the DP induces negative correlation between any two disjoint

sets, without taking into account the distance between them (Gelman et al. 2013, sec. 23.2). Therefore, the DP tends to have poor performance because it overly adheres to the empirical frequencies, as illustrated in Figure 1. This behavior can also be appreciated from the posterior mean of $P(A)$, that is, the predictive law $P_n(A) = \mathbb{E}\{P(A) | y_{1:n}\} = \mathbb{P}(Y_{n+1} \in A | y_{1:n})$, which is

$$P_n(A) = \frac{\alpha}{\alpha + n} P_0(A) + \frac{1}{\alpha + n} \sum_{i=1}^n \mathbb{1}(y_i \in A), \quad (2)$$

for any measurable set $A \subseteq \mathcal{Y}$ and where $\mathbb{1}(\cdot)$ denotes the indicator function. It is reasonable to expect that a better estimator for count-data distributions could be obtained by replacing $\mathbb{1}(\cdot)$ in (2) with a nondegenerate kernel, thereby allowing for the borrowing of information. Early ideas along these lines can be found in Hjort (1994a, 1994b). Such a direct elicitation of the law $P_n(\cdot)$ is referred to as the *predictive approach*. This is advantageous because it allows one to directly specify simple predictive rules with desirable properties, such as smoothness. In contrast, within the traditional Bayesian nonparametric framework, it is not obvious how to define a nonparametric process P that preserves the simplicity and flexibility of the DP while still allowing smoothing.

In a predictive approach, the statistical uncertainty is taken into account by modeling the conditional distribution of new data Y_{n+1} given past observations $y_{1:n}$. For instance, Blackwell and MacQueen (1973) described the predictive rule resulting from the DP prior using a Pólya urn scheme for which $Y_1 \sim P_0$ and $Y_{n+1} | y_{1:n} \sim P_n$ for $n \geq 1$, with $P_n(A) = \mathbb{P}(Y_{n+1} \in A |$

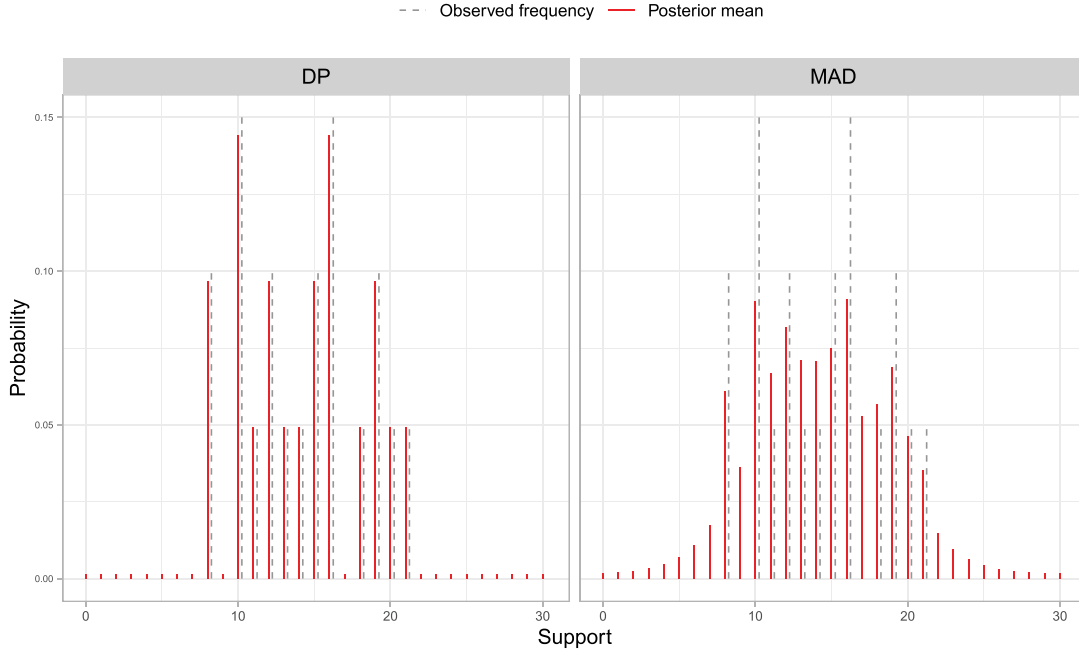


Figure 1. Posterior mean $P_n(Y = y) = \mathbb{E}\{P(Y = y) \mid y_{1:n}\}$, with $y \in \{0, \dots, 30\}$, obtained using a DP prior (left) and our MAD sequence (right) on a simulated dataset with $n = 20$.

$y_{1:n}$ defined in (2). The sequence $(Y_n)_{n \geq 1}$ generated according to (2) is known as a Dirichlet sequence and is a conditionally iid random sample from P , with $P \sim \text{DP}(\alpha, P_0)$. Although predictive laws can be obtained by marginalizing models of the form in (1), potentially with alternative priors used in place of the DP, we instead start by defining the predictive law P_n . Thanks to Ionescu-Tulcea's theorem one can characterize the law of the joint sequence $(Y_n)_{n \geq 1}$ by specifying a sequence of one-step-ahead predictive distributions $(P_n)_{n \geq 0}$. Moreover, if the resulting sequence is exchangeable, de Finetti's theorem implies the existence of a random probability measure P , conditionally on which the data are iid (see Fortini and Petrone 2012; Berti et al. 2025). This strategy bypasses prior elicitation; in our case, avoiding the difficult problem of directly specifying a prior for P . Moreover, it highlights how the tasks of inference and prediction are intrinsically connected within the Bayesian paradigm. There is a growing recent literature discussing the foundations of the predictive approach; for a comprehensive review, we refer to Fortini and Petrone (2025). For further discussion on future directions of the predictive approach in Bayesian statistics, see also Fortini and Petrone (2023) and Holmes and Walker (2023).

While Fortini, Ladelli, and Regazzini (2000) provide sufficient conditions for a predictive distribution to generate an exchangeable sequence, these conditions are typically difficult to satisfy in practice. For this reason, recent works have focused on generalizing the Bayesian predictive framework by relaxing exchangeability to the weaker requirement that the sequence $(Y_n)_{n \geq 1}$ is conditionally identically distributed (CID; Kallenberg 1988; Berti, Pratelli, and Rigo 2004). This still guarantees the existence of a (possibly implicit) prior over P , with exchangeability recovered asymptotically. The literature about CID sequences (e.g., Berti, Pratelli, and Rigo 2004, 2012; Fortini, Petrone, and Sporysheva 2018; Berti et al. 2021, 2023a, 2025) has strong roots in probability theory, but the statistical properties of such

sequences have not been fully investigated, with a few notable exceptions. An approach based on the direct specification of copulas is discussed in Hahn, Martin, and Walker (2018); Fong, Holmes, and Walker (2023), whose focus is on continuous data with a Gaussian copula; see also Cui and Walker (2024), Fong and Yiu (2024b), and Bissiri and Walker (2025) for related works. Such an update is designed for continuous data, and when modeling multivariate distributions, it is dimension ordering dependent. As an alternative, Fortini and Petrone (2020) provided a Bayesian interpretation for the popular Newton recursive density estimator (Newton and Zhang 1999; Newton 2002) and proved that the resulting sequence is CID. Although, in principle, the Newton estimator can be adapted to the discrete case, the algorithm requires a numerical evaluation of an integral at each step, which could lead to computational bottlenecks if a closed form is not available. Unfortunately, although replacing $\mathbb{1}(\cdot)$ in (2) with a nondegenerate kernel yields a fast and accurate estimator, the corresponding sequence $(Y_n)_{n \geq 1}$ is neither exchangeable (Sariev and Savov 2024) nor CID for a general choice of kernel; see also Berti et al. (2023b) for a rigorous discussion.

In this article, we introduce a novel generalized Bayesian predictive method to model count data distributions. We propose the recursive predictive rule

$$P_n(A) = (1 - w_n)P_{n-1}(A) + w_n K_n(A \mid y_n), \quad (3)$$

for any measurable set $A \subseteq \mathcal{Y}$, where $(w_n)_{n \geq 1}$ is a sequence of decreasing weights $w_n \in (0, 1)$, and, for all $n \geq 1$ we let $K_n(\cdot \mid y_n) = K(\cdot; y_n, P_{n-1})$ be a sequence of Metropolis-Hastings (MH) kernels, which will be formally defined in Section 2. Intuitively, the MH kernel is a modification of a smoothing kernel where the probability mass is redistributed to satisfy the CID condition. The special case (2) is recovered for $w_n = (\alpha + n)^{-1}$ and $K_n(\cdot \mid y_n) = \mathbb{1}(y_n \in \cdot)$. The simple structure (3) resembles

a frequentist kernel density estimator (KDE, Wand and Jones 1995) under a particular choice of weights w_n , where MH kernels centered at the observed data points serve a similar role to local kernels in KDE. More precisely, if $w_n = (\alpha + n)^{-1}$, then the recursion in (3) becomes

$$P_n(A) = \frac{\alpha}{\alpha + n} P_0(A) + \frac{1}{\alpha + n} \sum_{i=1}^n K_i(A | y_i).$$

This is similar in spirit to the recursive predictive algorithms of Hahn, Martin, and Walker (2018) and Fong, Holmes, and Walker (2023), but, crucially, we specify a kernel $K_n(\cdot | y_n)$ tailored for count data. The predictive rule (3) with MH kernels generates a Metropolis-adjusted Dirichlet (MAD) sequence, which is a CID sequence admitting a recursive structure (Berti et al. 2023a). Consequently, MAD sequences are asymptotically exchangeable, which means that, for large n , $(Y_n)_{n \geq 1}$ can be regarded as an approximately iid sample from a random distribution P . In other words, the predictive (3) implies the existence of an implicit prior for P , which is a novel alternative to a DP. The posterior of P can be regarded as a martingale posterior in the sense of Fong, Holmes, and Walker (2023), implying that a predictive resampling algorithm can be used to obtain posterior samples for P allowing uncertainty quantification.

We argue that employing MH kernels is particularly convenient for modeling count distributions for several reasons. First, these kernels offer a clear interpretation as a mixture of an arbitrary kernel centered on past data and the DP update, which is recovered as a special case. This structure enables the borrowing of information between nearby values in the support when updating the predictive rule, resulting in a smoother estimator than the DP. Second, computing the MAD predictive is easy in practice, with (3) representing the posterior mean of P . Consequently, obtaining a point estimate is efficient and does not require predictive resampling nor MCMC. Third, the properties of the kernel facilitate extensions to the multivariate case and nonparametric regression, as the update remains invariant to the ordering of dimensions. This approach is also suitable for modeling multivariate binary data and proves particularly effective for nonparametric regression with binary covariates, as demonstrated through simulations and real data analyses. Moreover, we obtain an asymptotic Gaussian approximation for the posterior distribution of P , highlighting the roles of the weights w_n and the kernel bandwidth in the learning mechanism.

2. Metropolis-Adjusted Dirichlet Sequences

We introduce a novel nonparametric predictive framework for modeling count data. Our goal is to develop a predictive construction that preserves the simplicity and flexibility of the Pólya urn scheme, while allowing for borrowing of information. Let $(Y_n)_{n \geq 1}$ be a sequence of random variables defined on a countable space $\mathcal{Y}$. In the proposed approach, we assume that

$$Y_1 \sim P_0, \quad \text{and} \quad Y_{n+1} | y_{1:n} \sim P_n, \quad (4)$$

for $n \geq 1$, with $P_n(\cdot) = \mathbb{P}(Y_{n+1} \in \cdot | y_{1:n})$ defined as in (3) for some sequence of decreasing weights $(w_n)_{n \geq 1}$. In practice, we require $\sum_{n \geq 1} w_n = \infty$ and $\sum_{n \geq 1} w_n^2 < \infty$ as this ensures good frequentist properties (Fortini and Petrone 2020; Fong,

Holmes, and Walker 2023), although this is not necessary to guarantee that $(Y_n)_{n \geq 1}$ is CID. The weights $w_n = (\alpha + n)^{-\lambda}$, with $\alpha > 0$ and $\lambda \in (0.5, 1]$, satisfy the required conditions. Importantly, this sequential construction leads to a unique and well-defined joint distribution $\mathbb{P}$ for the entire sequence $(Y_n)_{n \geq 1}$ thanks to the celebrated Ionescu–Tulcea theorem (e.g., Berti et al. 2025), which serves as the mathematical foundation of the predictive approach. To streamline the notation, we use capital letters P_0 , $P_n(\cdot)$, and $K_n(\cdot | y_n)$ to denote probability measures, and lowercase $p_0(y)$, $p_n(y)$, and $k_n(y | y_n)$ for their corresponding probability mass functions. This convention will be used consistently throughout the article.

Definition 1. Let $\mathcal{Y}$ be a countable space and consider the recursive predictive rule (3) for $n \geq 1$

$$p_n(y) = (1 - w_n)p_{n-1}(y) + w_n k_n(y | y_n), \quad y \in \mathcal{Y},$$

with initial distribution $p_0(y)$ and weight $w_n \in (0, 1)$. The Metropolis-Hastings kernel is defined as

$$k_n(y | y_n) = \gamma_n(y, y_n) k_*(y | y_n) + \mathbb{1}(y = y_n) \left[1 - \sum_{z \in \mathcal{Y}} \gamma_n(z, y_n) k_*(z | y_n) \right], \quad (5)$$

where $k_*(y | y_n)$ is a discrete base kernel and $\gamma_n(y, y_n)$ is a probability weight defined as

$$\gamma_n(y, y_n) = \gamma(y, y_n, p_{n-1}) = \min \left\{ 1, \frac{p_{n-1}(y) k_*(y_n | y)}{p_{n-1}(y_n) k_*(y | y_n)} \right\}. \quad (6)$$

The predictive distribution p_n is referred to as the Metropolis-adjusted Dirichlet (MAD) distribution with weights $(w_n)_{n \geq 1}$, base kernel k_* , and initial distribution p_0 . The corresponding sequence of random variables $(Y_n)_{n \geq 1}$, as defined in (4), is called a MAD sequence.

The kernel probability mass function in (5) is a mixture of a nondegenerate base kernel k_* centered at the observed data point and a point mass at y_n . If the base kernel is degenerate, that is, $k_*(y | y_n) = \mathbb{1}(y = y_n)$, then the acceptance probability $\gamma(y, y_n) = 1$ for all choices of y and y_n , implying that $k_n(y | y_n) = \mathbb{1}(y = y_n)$. The DP predictive distribution is then recovered if $w_n = (\alpha + n)^{-1}$. In general, the base kernel $k_*(y | y_n)$ is weighted by $\gamma_n(y, y_n)$, and the remaining probability mass $r_n(y_n) = 1 - \sum_{z \in \mathcal{Y}} \gamma_n(z, y_n) k_*(z | y_n)$ is assigned to y_n . When $r_n(y_n) = 0$ for a given y_n , there is no modification of the base kernel, and we have $k_n(y | y_n) = k_*(y | y_n)$ for all $y \in \mathcal{Y}$. This occurs when k_* is symmetric, that is, $k_*(y | y_n) = k_*(y_n | y)$, and $p_{n-1}(y) \geq p_{n-1}(y_n)$ for all $y \in \mathcal{Y}$. Informally, this means that if the new observation y_n lies in a region of $\mathcal{Y}$ to which p_{n-1} assigns low probability and k_* is symmetric, then the adjustment of the base kernel is negligible and $k_n(y | y_n) \approx k_*(y | y_n)$. In other words, the less information on y_n that is available, that is, when $p_{n-1}(y_n)$ is low, the more the MAD predictive borrows strength from nearby values.

2.1. Bayesian Properties of MAD Sequences

We provide a formal Bayesian justification for using a MAD sequence to model discrete data. Although MAD sequences are

not exchangeable, they are conditionally identically distributed (CID). The notion of CID sequences was introduced by Kallenberg (1988) as a relaxation of exchangeability, and it has been further developed by Berti, Pratelli, and Rigo (2004). A sequence $(Y_n)_{n \geq 1}$ of random variables is said to be CID if $\mathbb{P}(Y_n \in \cdot) = P_0(\cdot)$ for every $n \geq 1$, meaning that each Y_n has the same marginal distribution P_0 a priori, and

$$\mathbb{P}(Y_{n+t} \in \cdot \mid y_{1:n}) = \mathbb{P}(Y_{n+1} \in \cdot \mid y_{1:n}) = P_n(\cdot), \quad \text{for all } t \geq 1, n \geq 1. \quad (7)$$

It is sufficient to verify this condition for $t = 2$ in order to ensure its validity for all $t \geq 1$. Equivalently, Fong, Holmes, and Walker (2023) express condition (7) in a way that emphasizes the martingale property of the predictive distributions $(P_n)_{n \geq 1}$:

$$\mathbb{E}\{P_{n+1}(\cdot) \mid y_{1:n}\} = P_n(\cdot), \quad n \geq 1. \quad (8)$$

Note that $\mathbb{E}\{P_{n+1}(\cdot) \mid y_{1:n}\} = \mathbb{P}(Y_{n+2} \in \cdot \mid y_{1:n})$, which highlights the equivalence between the martingale and CID conditions. See Berti, Pratelli, and Rigo (2012, Theorem 3.1) for a formal proof.

The CID assumption replaces exchangeability, providing a formal Bayesian justification, as we will discuss below. A natural first candidate for the predictive update would be $p_n(y) = (1 - w_n)p_{n-1}(y) + w_n k_*(y \mid y_n)$, namely replacing the indicator function in the DP predictive in (2) with a nondegenerate base kernel. However, the resulting sequence of random variables $(Y_n)_{n \geq 1}$ would generally be neither exchangeable nor CID. Thus, the MH kernel can be viewed as a necessary adjustment of the base kernel to satisfy the martingale property. Specifically, while $\mathbb{E}\{k_*(y \mid Y_{n+1}) \mid y_{1:n}\} \neq p_n(y)$, the detailed balance property of the MH kernel ensures that $\mathbb{E}\{k_{n+1}(y \mid Y_{n+1}) \mid y_{1:n}\} = p_n(y)$ for all $y \in \mathcal{Y}$ and $n \geq 0$, which is precisely what is needed to prove the CID property (Berti et al. 2023a).

Theorem 1. Let $(Y_n)_{n \geq 1}$ be a MAD sequence. Then, for every choice of weight sequence $(w_n)_{n \geq 1}$, discrete base kernel k_* , and initial distribution p_0 , the sequence $(Y_n)_{n \geq 1}$ is CID.

Theorem 1 follows as a special case of Berti et al. (2023a, Theorem 2), since MAD sequences fall within the broad class of predictive updates based on stationary kernels. Indeed, the key property that we need to ensure is that $\mathbb{E}\{k_{n+1}(y \mid Y_{n+1}) \mid y_{1:n}\} = p_n(y)$, which is satisfied by any transition kernel $k_{n+1}(y \mid y_n)$ having $p_n(y)$ as its stationary distribution. For example, one could alternatively consider a MH kernel with the following acceptance probabilities:

$$\gamma_n(y, y_n) = g(\rho_n(y, y_n)), \quad \rho_n(y, y_n) = \frac{p_{n-1}(y)k_*(y_n \mid y)}{p_{n-1}(y_n)k_*(y \mid y_n)},$$

where $g : \mathbb{R}^+ \rightarrow (0, 1)$ is a function satisfying $g(t) = tg(1/t)$, which includes $g(t) = \min(1, t)$, the canonical choice, and $g(t) = t/(1 + t)$ as special cases. Any member of this class preserves the detailed balance condition (Tierney 1998), ensures stationarity of the kernel, and hence leads to a CID sequence. Throughout the article, we will focus on the canonical case $g(t) = \min(1, t)$.

Let $\theta = P(f) = \sum_{y \in \mathcal{Y}} f(y)p(y)$ denote any functional of interest, and analogously let $\theta_n = P_n(f) = \sum_{y \in \mathcal{Y}} f(y)p_n(y)$, for $n \geq 0$, denote its predictive counterpart.

Corollary 1 (Berti, Pratelli, and Rigo 2004). Consider a MAD sequence $(Y_n)_{n \geq 1}$. Then, $\mathbb{P}$ -a.s.,

(a) The sequence is asymptotically exchangeable, that is

$$(Y_{n+1}, Y_{n+2}, \dots) \xrightarrow{d} (Z_1, Z_2, \dots), \quad n \rightarrow \infty,$$

where $(Z_1, Z_2, \dots)$ is an exchangeable sequence with directing random probability measure P ;

(b) The corresponding sequence of predictive distributions $(P_n)_{n \geq 1}$ and empirical distributions $(\hat{P}_n)_{n \geq 1}$, where $\hat{P}_n = n^{-1} \sum_{i=1}^n \delta_{Y_i}$, converge weakly to the random probability measure P ;

(c) For every $n \geq 1$ and every integrable function $f : \mathcal{Y} \rightarrow \mathbb{R}$, we have $\mathbb{E}(\theta) = \theta_0$ and $\mathbb{E}(\theta \mid y_{1:n}) = \theta_n$.

These results ensure that MAD sequences are a valid Bayesian method for nonparametric inference, bypassing direct prior specification on the space of probability distributions. The results of Corollary 1 are based on the general results provided by Berti, Pratelli, and Rigo (2004) for CID sequences. Although the probability law implied by a MAD sequence is not ordering-invariant, part (a) of Corollary 1 guarantees that this dependence vanishes asymptotically. As a result, for large n , exchangeability is recovered and, informally, we will say $Y_n \mid P \stackrel{\text{iid}}{\sim} P$. Thus, an asymptotic equivalent of de Finetti's theorem holds: each MAD sequence has a corresponding unique prior on P and part (b) guarantees that P is defined as the limit of the predictive distributions. This convergence holds in the weak sense, but it can be strengthened to total variation convergence (see Lemma S1 in supplementary material S1). Finally, by part (c) we have $\theta = \mathbb{E}\{P(f)\} = P_0(f) = \theta_0$ for every integrable function f , so that P_0 retains the role of a base measure as for standard Dirichlet sequences, providing an initial guess at P . In particular, if $f(y) = \mathbb{1}_A(y)$ is the indicator function for some measurable $A \subseteq \mathcal{Y}$, then we get $\mathbb{E}\{P(A)\} = P_0(A)$. Consequently, prior or external information can still be incorporated into the model through P_0 . Theorem 1 and Corollary 1 are valid for a general sequence of weights $(w_n)_{n \geq 1}$. When $w_n = (\alpha + n)^{-\lambda}$, the parameter α retains the same interpretation as the precision parameter in the DP, controlling the degree of shrinkage of P toward P_0 . Setting large values for α denotes a strong confidence in the base measure, while $\alpha = 1$ provides a useful weakly informative default value. For a discussion of the role of λ based on asymptotic considerations, we refer to Section 2.4.

Given the data $y_{1:n}$, the predictive distributions P_n converge to the posterior distribution for P , which can be considered a martingale posterior in the sense of Fong, Holmes, and Walker (2023). Prior and posterior sampling of P can be performed using a simple Monte Carlo scheme (see Section 2.3). To illustrate, if the sequence of predictive distributions is (2), then the output of the posterior sampling algorithm for a given set $A \subseteq \mathcal{Y}$ would be an approximate iid sample from $P(A) \mid y_{1:n}$, whose distribution is

$$\text{Beta}\left(\alpha P_0(A) + \sum_{i=1}^n \mathbb{1}(y_i \in A), \alpha + n - \alpha P_0(A) - \sum_{i=1}^n \mathbb{1}(y_i \in A)\right),$$

that is, the distribution of $P(A) \mid y_{1:n}$ when assuming a DP prior. However, to obtain a point estimate for any functional of interest $\theta = P(f)$, sampling is not necessary. By part (c), the posterior mean is $\mathbb{E}(\theta \mid y_{1:n}) = \mathbb{E}\{P(f) \mid y_{1:n}\} = P_n(f) = \theta_n$, which is a consequence of the martingale property.

Although the posterior variance of $P(f)$ depends on the unknown limit distribution P , examining the variance of $P_{n+1}(f)$ given the observed data provides useful insights into MAD sequences. The expected variation in the predictive distribution after observing a new data point is expressed as

$$\begin{aligned} \text{var}\{P_{n+1}(f) \mid y_{1:n}\} &= \mathbb{E}\{[P_{n+1}(f) - P_n(f)]^2 \mid y_{1:n}\} \\ &= w_{n+1}^2 \left\{ \sum_{y \in \mathcal{Y}} K_n(f \mid y) p_n(y) - P_n(f)^2 \right\}. \end{aligned} \quad (9)$$

When no data are observed, (9) represents the expected prior-to-posterior variability after the first observation. We will show in Section 2.4 that the variance of P_{n+1} is also closely related to the asymptotic variance of P . Moreover, the correlation between two measurable functions f_1, f_2 induced by P_{n+1} given $y_{1:n}$ is

$$\begin{aligned} \text{cor}\{P_{n+1}(f_1), P_{n+1}(f_2) \mid y_{1:n}\} &= \frac{\sum_{y \in \mathcal{Y}} K_n(f_1 \mid y) K_n(f_2 \mid y) p_n(y) - P_n(f_1) P_n(f_2)}{\sqrt{\left[\sum_{y \in \mathcal{Y}} K_n(f_1 \mid y) p_n(y) - P_n(f_1)^2 \right] \left[\sum_{y \in \mathcal{Y}} K_n(f_2 \mid y) p_n(y) - P_n(f_2)^2 \right]}}. \end{aligned} \quad (10)$$

Unlike the DP, MAD sequences can induce positive correlations between the probabilities of nearby support values for suitable choices of the base kernel, which makes them especially appealing in applications. The variance and correlation of P_{n+1} given $y_{1:n}$ under a specific base kernel are examined in detail in Section 2.2.

Remark 1. In general, the predictive law P_n of a CID sequence depends on the order of the observations $y_{1:n}$, which is typically unappealing. In contrast, the predictive law P_n of an exchangeable sequence does not depend on the order of $y_{1:n}$ (Fortini, Ladelli, and Regazzini 2000). This raises the question of whether a CID sequence whose predictive distribution is order-invariant even exists. While order-invariant CID sequences are theoretically possible, a profound and elegant answer is provided in Fortini, Petrone, and Sporysheva (2018, Corollary 3.7), which establishes that a CID sequence is exchangeable if and only if its predictive law is symmetric with respect to the conditioning events. In other words, some form of asymmetry in martingale posteriors is unavoidable—otherwise, we would revert to the classical exchangeable framework, which is often too restrictive.

2.2. On the Choice of the Base Kernel

In this section, we consider the univariate space $\mathcal{Y} = \{0, 1, \dots\}$, whereas in Section 3 we will address the multivariate case. Although the results in Section 2.1 hold for an arbitrary discrete base kernel k_* , the choice of kernel has important consequences for the MH update. For example, using Poisson base kernels—where a single parameter governs both the location and the scale—limits the flexibility of the MH kernel, as the same parameter simultaneously controls both the bandwidth and the center

of the distribution. A simple alternative, resembling common choices in KDE, is to consider a uniform kernel over the interval $\{y_n - m, \dots, y_n + m\}$ for some $m \geq 0$, that is, $k_*(y \mid y_n, m) = (2m + 1)^{-1}$ if $y \in \{y_n - m, \dots, y_n + m\}$ and 0 otherwise. Then, the resulting MH kernel is centered at y_n , with bandwidth controlled by m . The corresponding probability mass function becomes

$$\begin{aligned} k_n(y \mid y_n, m) &= \min \left\{ 1, \frac{p_{n-1}(y)}{p_{n-1}(y_n)} \right\} k_*(y \mid y_n, m) \\ &+ \mathbb{1}(y = y_n) \left[1 - \frac{1}{2m + 1} \sum_{z=y_n-m}^{y_n+m} \min \left\{ 1, \frac{p_{n-1}(z)}{p_{n-1}(y_n)} \right\} \right], \end{aligned}$$

for every $n \geq 1$, which has finite support $\{y_n - m, \dots, y_n + m\}$. When $m = 0$, the DP predictive distribution is recovered. Another simple possibility, paralleling classical choices in MCMC, is to choose a base kernel $k_*(y \mid y_n) = k_*(y)$ that does not depend on past observations, with $k_*(y)$ being a known distribution such as a Poisson or a negative binomial. This specification enforces a strong parametric structure and is suitable only when small deviations from a known model are expected. Although these two proposals lead to substantial computational simplifications, using more flexible kernels can improve predictive accuracy.

As an alternative, we recommend using a rounded Gaussian base kernel as in Canale and Dunson (2011), whose probability mass function is given by

$$k_*(y \mid y_n, \sigma) = \frac{\int_{y-1/2}^{y+1/2} \mathcal{N}(t \mid y_n, \sigma^2) dt}{\sum_{z \in \mathcal{Y}} \int_{z-1/2}^{z+1/2} \mathcal{N}(t \mid y_n, \sigma^2) dt},$$

where $\mathcal{N}(\cdot \mid y_n, \sigma^2)$ denotes a normal density function with mean y_n and variance σ^2 . While σ controls the bandwidth of the rounded Gaussian base kernel k_* , its role within the MH kernel k_n is less immediate to interpret, as a consequence of the MH correction. Figure 2 illustrates the behavior of k_n as a function of σ in a toy example. As the variance of the base kernel shrinks, that is, as $\sigma \rightarrow 0$, we get $k_*(y \mid y_n) = k_n(y \mid y_n) = \mathbb{1}(y = y_n)$ and the rejection rate $r_n(y_n) = 0$, yielding a DP-like predictive distribution as in (2) with a general set of weights $(w_n)_{n \geq 1}$, which fails at borrowing information. Conversely, for large values of σ the base kernel converges to a uniform distribution on the support $\mathcal{Y}$. If the support is finite, $|\mathcal{Y}| < \infty$, the rejection rate tends to $r_n(y_n) = 1 - 1/|\mathcal{Y}| \sum_{z \in \mathcal{Y}} \min\{1, p_{n-1}(z)/p_{n-1}(y_n)\}$ as $\sigma \rightarrow \infty$, which is low whenever $p_{n-1}(y_n)$ is. In this limiting case, the borrowing of information is maximal and distributed uniformly across support points when y_n has low probability under p_{n-1} , and minimal, essentially collapsing to a point mass at y_n , when $p_{n-1}(y_n)$ is large. Intermediate values of σ therefore yield a less dichotomic borrowing of information across nearby probabilities. The role of σ in MAD sequences is therefore similar, though not identical, with that in classical kernel density estimators.

The bandwidth σ also impacts the posterior variability as well as the correlation structure. As discussed in Section 2.1, we focus on the variance and correlation of $P_{n+1} \mid y_{1:n}$, since the limit distribution P is unknown. The left panel of Figure 3

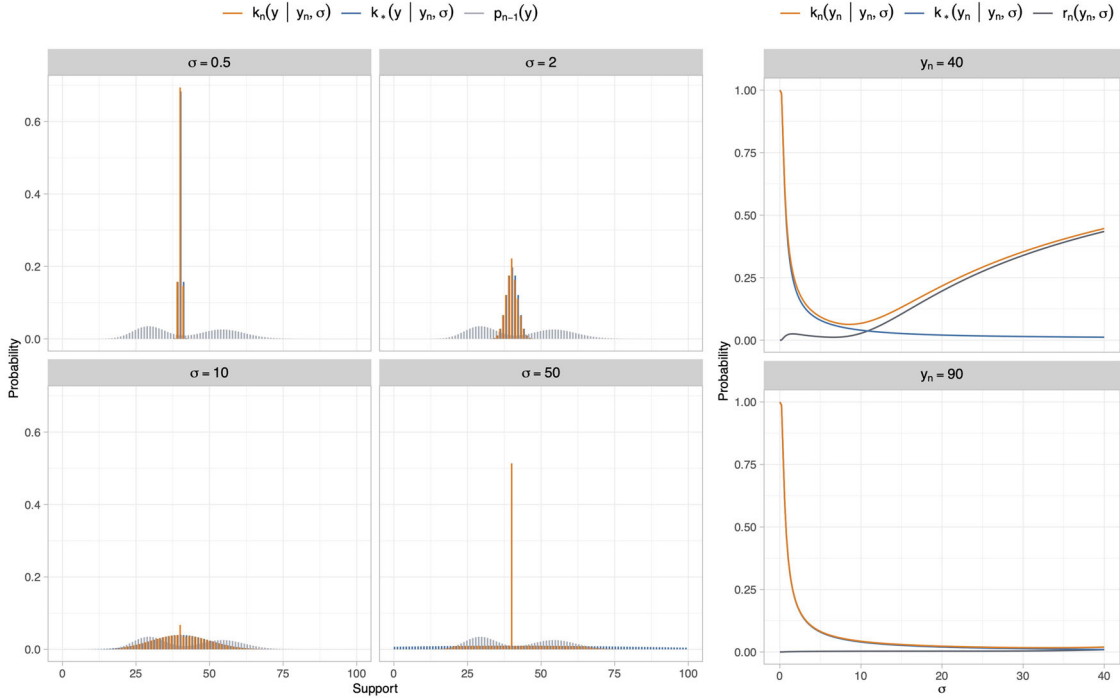


Figure 2. Left and center: examples of the probability mass function of the MH kernel using a rounded Gaussian base kernel with $\sigma \in \{0.5, 2, 10, 50\}$ after observing $y_n = 40$. Right: values of $k_n(y_n | y_n, \sigma)$, $k_*(y_n | y_n, \sigma)$, and $r_n(y_n, \sigma)$ as functions of σ for $y_n \in \{40, 90\}$, with $\mathcal{Y} = \{0, \dots, 200\}$. Although k_* is the same for both choices of y_n , the behavior of $r_n(40, \sigma)$ and $r_n(90, \sigma)$ differs substantially as $\sigma \rightarrow \infty$, because $p_{n-1}(40) > p_{n-1}(90)$.

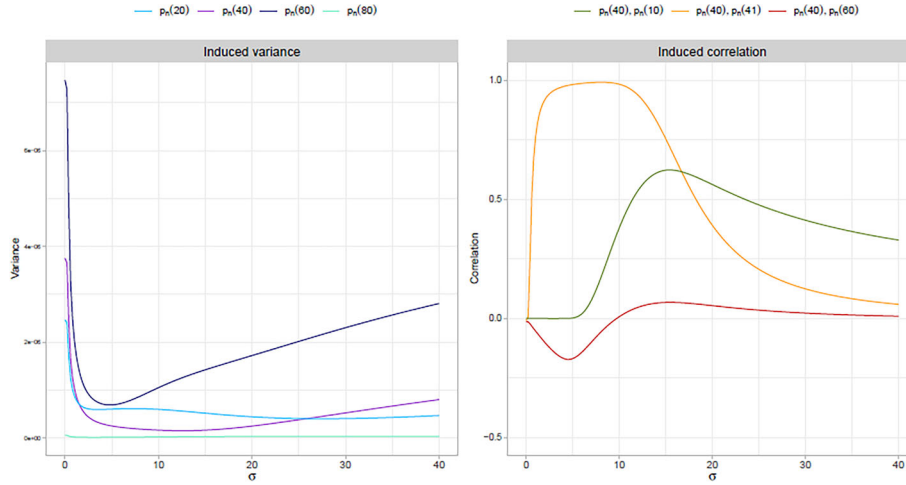


Figure 3. Left: posterior variance of $p_n(y)$ given $y_{1:n-1}$, for $y \in \{20, 40, 60, 80\}$. Right: correlation between $p_n(40)$ and $p_n(x)$ given $y_{1:n-1}$, for $x \in \{10, 41, 60\}$. In both panels, p_{n-1} is the same as in Figure 2.

illustrates how σ influences the variance of $P_{n+1} | y_{1:n}$ in the same toy example of Figure 2. For $\sigma = 0$, the DP is recovered, and the variance is maximum. In this case, (9) is proportional to the posterior variance induced by the DP. As σ grows, the MH kernel becomes more spread-out and the variance dramatically decreases. The variance increases again for large values of σ , corresponding to concentrated MH kernels as discussed above. Finally, unlike in the DP case, the induced correlation between any pair of disjoint sets A_1, A_2 for MAD sequences is not restricted to be negative, as illustrated in the right panel of Figure 3. This is a very appealing feature, because positive correlation enables borrowing of information between nearby values.

2.3. Uncertainty Quantification via Predictive Resampling

A predictive resampling algorithm (Fortini and Petrone 2020; Fong, Holmes, and Walker 2023), described in Algorithm 1, can be used for approximate posterior sampling of P , whose law, unfortunately, is generally not available in closed form. Algorithm 1 can also be used for prior sampling of P by setting $n = 0$. Starting from P_n , the algorithm performs forward simulations by recursively sampling and updating the MAD predictive distribution. Each forward simulation generates a random dataset $(Y_{n+1}, \dots, Y_N)$ which can be used to compute $P_N(\cdot)$ or any other statistic of interest, such as the population mean. By part (b) of Corollary 1, we can regard

Algorithm 1 Predictive resampling (Fortini and Petrone 2020; Fong, Holmes, and Walker 2023)

```

Compute  $p_n$  as in (3) from the observed data  $y_{1:n}$ 
Set  $N \gg n$ 
for  $j = 1, \dots, B$  do
  for  $i = n + 1, \dots, N$  do
    Sample  $Y_i \mid y_{1:i-1} \sim p_{i-1}$ 
    Update  $p_i(y) = (1 - w_i)p_{i-1}(y) + w_i k_i(y \mid y_i)$ 
  end for
end for
return  $p_N^{(1)}(y), \dots, p_N^{(B)}(y)$ , an iid sample of size  $B$  from the
distribution of  $p_N(\cdot) \mid y_{1:n}$ 

```

P_N as a finite-sample approximation of the limit distribution P for sufficiently large N . Consequently, the output of **Algorithm 1** provides an approximate sample from the distribution of $P(\cdot) \mid y_{1:n}$.

Clearly, it is also possible to perform posterior inference on any functional $\theta = P(f)$. For example, if one is interested in the population mean $\theta = P(f) = \sum_{y \in \mathcal{Y}} y p(y)$, then the corresponding martingale posterior distribution can be approximated by considering its finite-sample version $\theta_N = P_N(f) = \sum_{y \in \mathcal{Y}} y p_N(y)$ which can be easily simulated using **Algorithm 1**. However, as previously mentioned, the posterior mean does not require approximations nor sampling, as we have $\mathbb{E}(\theta \mid y_{1:n}) = \theta_n$ thanks to the martingale property.

Remark 2. The discreteness of the MAD predictive rule is computationally advantageous compared to the continuous case (e.g., Fong, Holmes, and Walker 2023). First, despite what the Metropolis–Hastings terminology might suggest, the recursive update of p_n does not involve any sampling, as p_n can be directly updated using the definition of the kernel $k_n(y \mid y_n)$. This update is straightforward when both p_0 and the base kernel k_* have finite support, since p_n can then be represented by a finite set of probability weights. Sampling from p_n is also straightforward, again because it involves a finite set of probabilities. In the countable case, it suffices to truncate p_n by removing support points with negligible probability, which is effectively equivalent to using a base kernel with finite support.

2.4. Asymptotic Behavior

In this section, we establish a predictive central limit theorem for MAD sequences. Specifically, for any $H \geq 1$ and any measurable sets $A_1, \dots, A_H$, we derive a Gaussian approximation for the posterior distribution of $P(A_1), \dots, P(A_H) \mid y_{1:n}$ as $n \rightarrow \infty$. Intuitively, our result quantifies the uncertainty that arises from observing a finite sample of size n instead of the infinite population. Within the Bayesian predictive literature, Fortini and Petrone (2020) provide a predictive central limit theorem for the Newton estimator, while Fong and Yiu (2024a) obtain a similar result for parametric martingale posteriors. The following results specialize those of Fortini and Petrone (2025) for general CID sequences; however, in the context of MAD sequences, some of their assumptions are not required, and therefore the proof is provided.

Proposition 1. Let $(Y_n)_{n \geq 1}$ be a MAD sequence and let $(r_n)_{n \geq 1}$ be a monotone sequence of positive numbers such that $r_n = (\sum_{k > n} w_k^2)^{-1}$ as $n \rightarrow \infty$. For every $n \geq 1$ and all measurable sets $A_1, \dots, A_H$, let $\Sigma_n = \Sigma_n(A_1, \dots, A_H) = [\Sigma_{n,jt}]$, $j, t = 1, \dots, H$, with

$$\Sigma_{n,jt} = \sum_{y \in \mathcal{Y}} K_{n+1}(A_j \mid y) K_{n+1}(A_t \mid y) p_n(y) - P_n(A_j) P_n(A_t). \quad (11)$$

Assume $\sqrt{r_n} \sup_{k \geq n} w_k \rightarrow 0$, $\sum_{k \geq 1} r_k^2 w_{k+1}^4 < \infty$, and that, for every $H \geq 1$ and all $A_1, \dots, A_H$, the random matrix $\Sigma = \Sigma(A_1, \dots, A_H) = [\Sigma_{jt}]$, $j, t = 1, \dots, H$, with

$$\Sigma_{jt} = \sum_{y \in \mathcal{Y}} K(A_j \mid y) K(A_t \mid y) p(y) - P(A_j) P(A_t),$$

where $K(A \mid y) = \sum_{x \in A} k(x \mid y)$ and

$$\begin{aligned} k(x \mid y) &= \gamma(x, y) k_*(x \mid y) \\ &\quad + \mathbb{1}(x = y) \left[1 - \sum_{z \in \mathcal{Y}} \gamma(z, y) k_*(z, y) \right], \\ \gamma(x, y) &= \min \left\{ 1, \frac{p(x) k_*(y \mid x)}{p(y) k_*(x \mid y)} \right\}, \end{aligned}$$

is positive definite. Then, $\mathbb{P}$ -a.s., $\Sigma_n \rightarrow \Sigma$ and

$$\begin{aligned} \sqrt{r_n} \Sigma_n^{-1/2} \begin{bmatrix} P(A_1) - P_n(A_1) \\ \vdots \\ P(A_H) - P_n(A_H) \end{bmatrix} \Big| y_{1:n} &\xrightarrow{d} \mathbf{Z}, \\ \mathbf{Z} &\sim \mathcal{N}_H(\mathbf{0}, \mathbf{I}), \end{aligned}$$

for $n \rightarrow \infty$, where $\mathcal{N}_H(\mathbf{0}, \Sigma)$ denotes a H -dimensional Gaussian distribution with mean $\mathbf{0}$ and covariance Σ , and $\mathbf{I}$ denotes the identity matrix.

Remark 3. The result of **Proposition 1** holds when $w_n = (\alpha + n)^{-\lambda}$ with $r_n = (2\lambda - 1)n^{2\lambda-1}$.

A more detailed discussion of the derivation of **Proposition 1** is provided in Section S1 of the supplementary material. The covariance estimator (11) is proportional to $\text{cov}\{P_{n+1}(A_j), P_{n+1}(A_t) \mid y_{1:n}\}$. The proposition provides insights on the relationship between the learning rate of the predictive rule and the posterior variability, highlighting the crucial role played by the weights w_n : while weights that decay to zero quickly induce fast learning and convergence to the asymptotic exchangeability regime, small values of w_n may result in poor learning and underestimation of the posterior variability. This was also noted by Fortini and Petrone (2020, 2025) and Fong and Yiu (2024a), which we refer to for an extensive discussion.

Although the standard choice for the DP is $w_n = (\alpha + n)^{-1}$, it has been observed that such weights may decay to zero too quickly according to frequentist criteria, making them unappealing for CID modeling (see Fortini and Petrone 2020; Fong, Holmes, and Walker 2023). To address this, the literature has proposed alternatives with slower decay rates, such as $w_n = (\alpha + n)^{-2/3}$ (Martin and Tokdar 2009), and $w_n = (2 - n^{-1})(n + 1)^{-1}$ (Fong, Holmes, and Walker 2023). Alternatively, Fortini

and Petrone (2020) suggest using adaptive weights of the form $w_n = (\alpha + n)^{-\lambda_n}$, where $\lambda_n = 1$ for $n \leq N_*$ to allow fast convergence, and $\lambda_n = 3/4$ when $n > N_*$ induces slower decay once an approximate exchangeability regime is reached. Although this split-sample approach is appealing, a smoother transition between the two regimes is desirable. For this reason, we consider the adaptive weights with

$$\lambda_n = \lambda + (1 - \lambda) \exp \left\{ -\frac{1}{N_*} n \right\}, \quad (12)$$

with $\lambda \in (1/2, 1]$ and $N_* > 0$. Using (12), the weights behave as $w_n \approx (\alpha + n)^{-1}$ for small n , smoothly transitioning to $w_n \approx (\alpha + n)^{-\lambda}$ as n increases. The transition rate is governed by N_* . Following Fortini and Petrone (2020), a practical default is to set $\lambda = 3/4$ and $N_* = 500$ in our analyses. The implications of these different weighting options for the MAD sequences are explored in Section 4 and in Section S4 of the supplementary material.

2.5. Univariate Illustrative Example

We provide a simulated example to illustrate the MAD sequence and compare it with the DP and the rounded Gaussian DP mixture (RG-DPM, Canale and Dunson 2011). We generate $n = 500$ data points from the mixture $0.3 \text{ Poi}(y; 1) + 0.2 \text{ Poi}(y; 18) + 0.3 \text{ Poi}(y; 40) + 0.3 \text{ Poi}(y; 62)$. We consider two variants of MAD sequences, having $w_n = (\alpha + n)^{-1}$ and $w_n = (\alpha + n)^{-\lambda_n}$, with λ_n defined in (12), $\lambda = 3/4$ and $N_* = 500$. For updating the MAD sequences, we choose a uniform base measure P_0 over the support, which we restrict to $\{0, \dots, 100\}$, and set $\alpha = 1$. We employ rounded Gaussian base kernels, and we recommend selecting σ using a data-driven approach. Following Fong, Holmes, and Walker (2023), we rely on the notion of prequential log-likelihood introduced by Dawid (1984). Specifically, under the predictive construction (4), the quantity $\sum_{i=1}^n \log p_{i-1}(y_i)$ can be interpreted as a log-likelihood function, making its maximization a natural criterion for selecting σ . Selecting the bandwidth that optimizes the prequential log-likelihood provides some robustness against overfitting, as this criterion is related to the cross-validation score (Fong and Holmes 2020) and evaluates how well the sequence of predictive distributions forecasts sequential data points. Alternatively, one can tune σ using the methods available for KDE, such as minimizing the prediction error in a validation set or by cross-validation.

Uncertainty quantification is carried out by predictive resampling $N = 10,000$ future observations starting from P_n and obtaining $B = 1000$ samples from the corresponding martingale posterior. By replacing exchangeability with the CID condition, MAD predictive distributions depend on the ordering of $y_{1:n}$. To ensure permutation invariance, we compute P_n averaging over S permutations of the data. Consistently with Fong, Holmes, and Walker (2023), we found that $S = 10$ is sufficient in practice. Computations for different permutations can be performed in parallel. The prequential log-likelihood for selecting the optimal σ is also averaged over the S permutations, whereas the predictive resampling algorithm starts from the permutation averaged P_n . When $w_n = (\alpha + n)^{-\lambda_n}$, choosing the optimal σ , computing the MAD predictive distribution and predictive resampling take 2.870, 0.096, and 10.735 sec, respectively, on a 3.2 GHz 8-Core Apple M1 CPU.

Figure 4 shows the predictive distributions obtained using the two variants of MAD sequences and DP. Both weight options provide similar MAD predictive distributions and the improvement over the DP is substantial. The MAD sequences effectively capture the four components of the mixture, assigning lower probabilities in the tails and in the region between the modes, while the RG-DPM fails to separate the two components of the mixture centered on 40 and 62. The RG-DPM takes 12.639 sec for sampling 6000 realizations from the posterior distribution, discarding the first half as burnin. Moreover, MAD sequences converge to the true distribution, as empirically shown in supplementary material S3. The top-right panel of Figure 4 presents the martingale posterior distribution for the quantity $\theta_N = P_N((-\infty, 25]) = \sum_{y \in \mathcal{Y}} \mathbb{1}(y \leq 25) p_N(y)$, computed as described in Section 2.3. To judge convergence of the predictive resampling algorithm for θ_N , we consider 100 sample paths of θ_m , $m = n + 1, \dots, 20,000$, as reported in the bottom-right panel of Figure 4. In this example, we see that $N = 10,000$ is sufficiently large for θ_N to approximate θ . Alternatively, Fong, Holmes, and Walker (2023) consider the L_1 distance between p_m and the initial p_n , which we include in Section S5 of the supplement. As discussed in Section 2.4, the adaptive weights yield a more conservative uncertainty quantification compared to the standard choice $w_n = (\alpha + n)^{-1}$. This results in wider 95% credible intervals for $P_N(y)$ and a less concentrated martingale posterior for θ .

3. Multivariate Modeling

In this section, we consider MAD sequences for modeling multivariate count data. Let $\mathbf{y}_n = (y_{n1}, \dots, y_{nd}) \in \mathcal{Y} = \{0, 1, \dots\}^d$ be a realization of the random vector $\mathbf{Y}_n = (Y_{n1}, \dots, Y_{nd})$, for $n \geq 1$ and $d \geq 2$, and denote $\mathbf{y}_{1:n} = (\mathbf{y}_1^\top, \dots, \mathbf{y}_n^\top)$. Assuming the sequential generating mechanism (4) for $(\mathbf{Y}_n)_{n \geq 1}$, the multivariate MAD predictive distribution has probability mass function

$$\begin{aligned} p_n(\mathbf{y}) &= \mathbb{P}(Y_{n+1,1} = y_1, \dots, Y_{n+1,d} = y_d \mid \mathbf{y}_{1:n}) \\ &= (1 - w_n) p_{n-1}(\mathbf{y}) + w_n k_n(\mathbf{y} \mid \mathbf{y}_n), \end{aligned}$$

where the multivariate MH kernel retains the same structure as in the univariate case, since the corresponding probability mass function is

$$\begin{aligned} k_n(\mathbf{y} \mid \mathbf{y}_n) &= \gamma_n(\mathbf{y}, \mathbf{y}_n) k_*(\mathbf{y} \mid \mathbf{y}_n) \\ &\quad + \mathbb{1}(\mathbf{y} = \mathbf{y}_n) \left[\sum_{\mathbf{z} \in \mathcal{Y}} \{1 - \gamma_n(\mathbf{z}, \mathbf{y}_n)\} k_*(\mathbf{z}, \mathbf{y}_n) \right], \end{aligned}$$

with

$$\gamma_n(\mathbf{y}, \mathbf{y}_n) = \min \left\{ 1, \frac{p_{n-1}(\mathbf{y}) k_*(\mathbf{y}_n \mid \mathbf{y})}{p_{n-1}(\mathbf{y}_n) k_*(\mathbf{y} \mid \mathbf{y}_n)} \right\}.$$

The above equations can be formally regarded as special cases of the general definition in (3), (5), and (6), but the multivariate nature of the data is emphasized using the bold notation. Any discrete multivariate distribution can be chosen as base kernel. A simple and reasonable choice is a factorized base kernel having rounded Gaussian distribution components,

$$k_*(\mathbf{y} \mid \mathbf{y}_n, \boldsymbol{\sigma}) = \prod_{j=1}^d k_*(y_j \mid y_{nj}, \sigma_j),$$

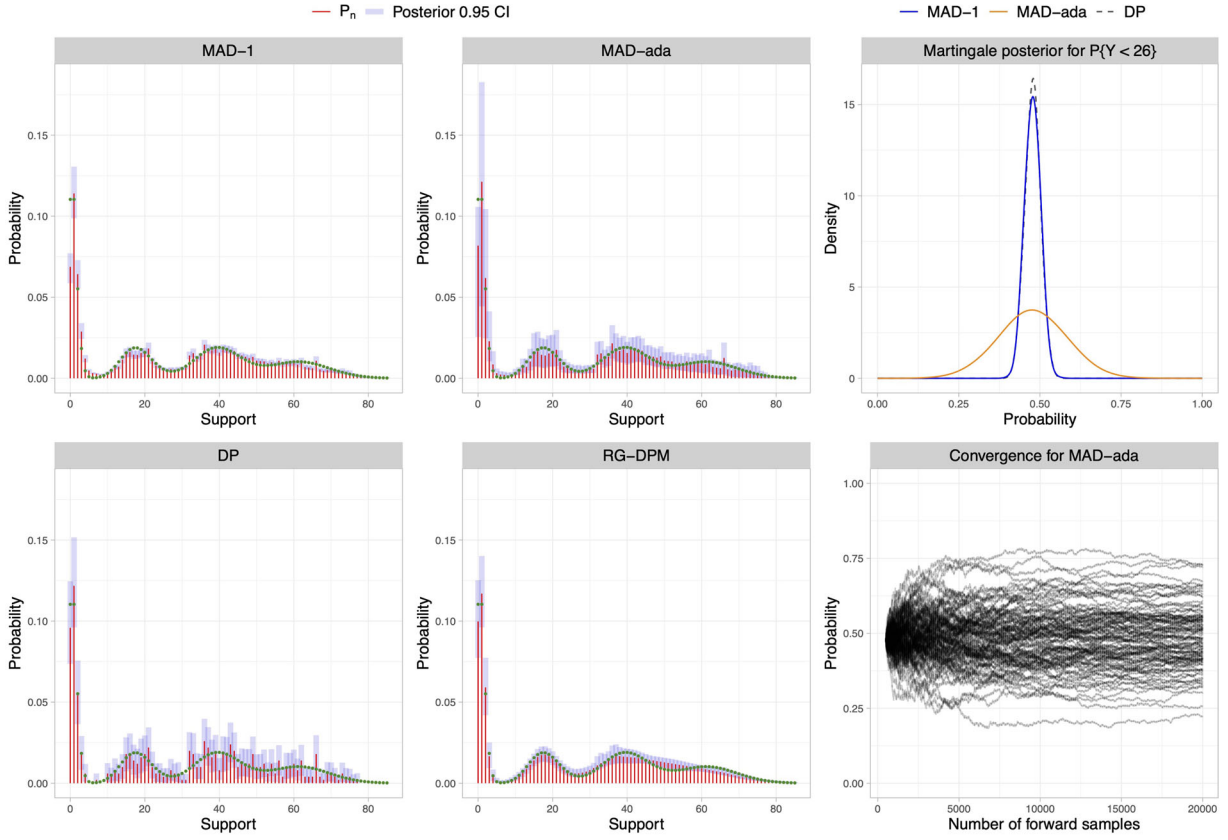


Figure 4. Predictive distributions from MAD-1, MAD-ADA, the DP, and the rounded Gaussian DP mixture (RG-DPM, Canale and Dunson (2011)). For each value $y \in \mathcal{Y}$ the 95% credible interval for $p_N(y)$ is included. The dotted line is the true probability mass function. Top-right: martingale posterior for $P_N((-\infty, 25])$. Bottom-right: 100 sample paths of $P_N((-\infty, 25])$ through forward sampling for MAD-ADA.

with $\sigma = (\sigma_1, \dots, \sigma_d)$. This factorized structure again parallels common practice in standard multivariate KDE. More complex kernels that induce dependence between components can also be adopted.

Remark 4. MAD sequences are defined on a general countable space $\mathcal{Y}$, which includes $\mathcal{Y} = \{0, 1, \dots\}^d$ as a special case. Hence, the results of Theorem 1 and Corollary 1, as well as those of Proposition 1, hold for multivariate MAD sequences. The multivariate case requires special attention, as the choice of the base kernel is less straightforward, and the practical implications deserve careful discussion.

There are several advantages of multivariate MAD sequences compared to existing approaches. First, although DP mixtures of multivariate rounded Gaussian kernels with non-diagonal covariance matrices (Canale and Dunson 2011) can be used to model multivariate count probability mass functions, MCMC computation can be inefficient. Conceptually, the predictive mixture model of Fortini and Petrone (2020) can address this issue, but in practice, the associated Newton algorithm requires the numerical evaluation of a multidimensional integral at each iteration. In contrast, employing MH kernels makes the extension to multivariate MAD sequences both straightforward and computationally tractable, as only univariate Gaussian integrals are involved, for which efficient numerical methods are available. Furthermore, unlike the multivariate method proposed by Fong, Holmes, and Walker (2023), which depends

on the order in which dimensions are updated, multivariate MAD sequences are invariant to the dimension ordering. For the copula update, the CID assumption holds only under a fixed ordering of the variables, and different orderings lead to distinct predictive distributions. This sensitivity is unappealing in practice. In contrast, MAD sequences assume that $(Y_n)_{n \geq 1}$ is a CID sequence regardless of the variable ordering. In supplementary material S2, we examine the dimension-ordering dependence of the copula update in a simulation study and compare the results with those obtained from MAD sequences.

3.1. Multivariate Binary Data

Multivariate MAD sequences can also be adapted to model multivariate binary data. Due to the flexibility of MH kernels, this extension is straightforward, requiring only the selection of an appropriate base kernel. For $y \in \{0, 1\}^d$, we propose to employ the base kernel that has probability mass function

$$k_*(y | y_n) = \prod_{j=1}^d k_*(y_j | y_{nj}),$$

where

$$k_*(y_j | y_{nj}) = |y_{nj} - \delta|^{y_j} (1 - |y_{nj} - \delta|)^{(1-y_j)} = \begin{cases} \delta^{y_j} (1 - \delta)^{1-y_j} & \text{if } y_{nj} = 0 \\ (1 - \delta)^{y_j} \delta^{1-y_j} & \text{if } y_{nj} = 1 \end{cases} \quad (13)$$

for $j = 1, \dots, d$ and $\delta \in [0, 0.5]$. An example of (13) for different values of δ is shown in supplementary material S3. Roughly speaking, for each dimension j , if $y_{nj} = 1$ the base kernel assigns probability $(1 - \delta) \geq 0.5$ to the event $Y_{n+1,j} = 1$ and nonzero probability $\delta \leq 0.5$ to the event $Y_{n+1,j} = 0$, and vice versa. For $\delta = 0$, the DP update is recovered, while for $\delta = 0.5$, a uniform base kernel is obtained. A practical default is to set $\delta = 0.25$, or δ can be estimated using the prequential log-likelihood technique in Section 2.2.

3.2. Nonparametric Regression and Classification

An important consequence of the multivariate approach is the possibility of using MAD sequences for nonparametric regression and classification. Let each covariate vector $\mathbf{x}_n \in \mathcal{X} \subseteq \{0, 1, \dots\}^d$ be a realization of the random vector $\mathbf{X}_n$. Then, we can first estimate the joint probability mass function $p_n(y, \mathbf{x})$ based on the joint CID sequence $(Y_1, \mathbf{X}_1), \dots, (Y_n, \mathbf{X}_n)$, and then compute the conditional predictive distribution $p_n(y | \mathbf{x})$, with $p_n(y | \mathbf{x}) = p_n(y, \mathbf{x}) / p_n(\mathbf{x})$. In this case, predictive resampling requires jointly sampling $(Y_{n+1}, \mathbf{X}_{n+1}) \sim p_n(y, \mathbf{x})$. The response variable Y_n can also be multivariate, possibly including both binary and count-valued components. Moreover, the framework can incorporate continuous covariates by combining discrete MH kernels with the bivariate Gaussian copula update proposed by Fong, Holmes, and Walker (2023). We provide simulation examples of count data regression and classification with binary covariates in Section 4.

4. Simulation Studies

In this section, we present two simulation studies comparing MAD sequences with established machine learning methods for count data regression and classification. The studies assess both the predictive accuracy and the quality of uncertainty quantification. A separate simulation comparing MAD sequences with the copula updating approach of Fong, Holmes, and Walker (2023) is provided in supplementary material S2.

4.1. Count Data Regression with Binary Covariates

Nonparametric methods are especially useful for capturing nonlinear relationships between variables. To demonstrate the effectiveness of MAD sequences in nonlinear regression settings, we conducted a simulation study comparing our approach with well-established machine learning methods, such as random forests and Bayesian additive regression trees (BART; Chipman, George, and McCulloch 2010). Although machine learning methods typically excel with large sample sizes, we expect that MAD sequences offer better performance in small-sample scenarios. As additional benchmarks, we include the DP and the Poisson generalized linear model (GLM). For each sample size $n \in \{40, 80\}$, we generate 50 simulated datasets. For each simulation, we include 10 binary covariates sampled from independent binomial distributions with parameter 0.5, and

$Y_i | \mathbf{x}_i \sim \text{Poi}(e^{\eta_i})$, where

$$\eta_i = 1 + \sqrt{|-0.5x_{i,1} + 1.5x_{i,2} + x_{i,3} + 0.5x_{i,4} - 0.5x_{i,5}|} + (-0.7x_{i,6} + 0.5x_{i,7} + 0.7x_{i,8} - 0.3x_{i,9} - 0.3x_{i,10})^2.$$

The study assesses the performance of each method by comparing the accuracy of out-of-sample predictions and uncertainty quantification. For predictive accuracy, we compute the mean squared error (MSE) obtained on 10^4 new observations. To evaluate uncertainty quantification accuracy, we consider the frequentist coverage of the 95% credible intervals for $\mathbb{E}(Y | \mathbf{x})$, calculated across 50 simulations for each of the $2^{10} = 1024$ possible values for the covariate vector $\mathbf{x}$. For random forests, we consider the predictive confidence intervals proposed by Mentch and Hooker (2016).

Since the posterior variability of MAD sequences depends on the choice of the weights w_n , we compare four variants of MAD sequences using the options discussed in Section 2.4: $w_n = (\alpha + n)^{-1}$ (MAD-1), $w_n = (\alpha + n)^{-2/3}$ (MAD-2/3), the weights $w_n = (2 - n^{-1})(n + 1)^{-1}$ proposed by Fong, Holmes, and Walker (2023) (MAD-DPM), and the adaptive solution $w_n = (\alpha + n)^{\lambda_n}$, with λ_n defined in (12) (MAD-ADA). We use a weakly informative uniform base measure over the support and set $\alpha = 1$. For the adaptive weights, we set $\lambda = 3/4$ and $N_* = 500$. Each MAD sequence employs a rounded Gaussian base kernel for the response, with bandwidth selected by maximizing the prequential log-likelihood. The hyperparameters for the binary base kernels for the covariates are set to the default value 0.25. To ensure permutation invariance, we average the predictive distribution and the prequential log-likelihood over 10 random permutations of the data. To compute the martingale posterior distributions we draw $B = 200$ samples from P_N using a predictive resampling algorithm with $N = n + 1000$. For point prediction with MAD sequences, we use the population mean $\theta_n = \sum_{y \in \mathcal{Y}} y p_n(y | \mathbf{x})$, while for uncertainty quantification, we employ the credible intervals obtained from the martingale posterior for the population mean, as described in Section 2.3.

The MSE and coverage results obtained across the 50 simulated datasets are reported in Table 1. Since it does not allow the borrowing of information between nearby locations, the DP overfits the observed data, consequently leading to poor out-of-sample point predictions and inaccurate credible intervals. In terms of predictive accuracy, as expected, MAD sequences perform substantially better than the Poisson GLM, which fails to capture nonlinear effects. The four variants of MAD sequences exhibit similar predictive performance, indicating that the choice of weights has little impact on the point estimate. Furthermore, all MAD sequences provide a better prediction accuracy than random forest and BART for both $n = 40$ and $n = 80$.

In terms of uncertainty quantification, the credible intervals produced by the MAD sequence with adaptive weights are, on average, well calibrated for both sample sizes considered. In contrast, alternative weighting strategies tend to result in undercoverage. However, we observe substantial variability in empirical coverage in $2^{10} = 1024$ possible covariate configurations. Although the Poisson GLM and BART models exhibit less variability in coverage across covariate settings, they consistently fall below the nominal level, indicating systematic undercoverage.

Table 1. Out-of-sample MSE evaluated on 10^4 new observations and median frequentist coverage for the 95% credible intervals for $\mathbb{E}(Y | \mathbf{x})$ for each of the 1024 possible values of the covariate vector $\mathbf{x}$.

Regression	$n = 40$				$n = 80$			
	MSE		Median coverage		MSE		Median coverage	
GLM	120.77	[51.51]	0.84	[0.70, 0.90]	94.93	[8.37]	0.90	[0.74, 0.96]
BART	101.17	[12.69]	0.92	[0.82, 0.96]	74.17	[10.00]	0.92	[0.84, 0.96]
RF	99.98	[7.45]	1.00	[0.92, 1.00]	87.75	[6.53]	1.00	[1.00, 1.00]
DP	1450.21	[5.53]	0.00	[0.00, 0.00]	1395.61	[8.72]	0.00	[0.00, 0.00]
MAD-1	91.07	[10.35]	0.92	[0.68, 0.98]	73.96	[7.60]	0.88	[0.58, 0.98]
MAD-2/3	88.83	[13.00]	0.92	[0.50, 0.98]	73.18	[9.58]	0.94	[0.56, 1.00]
MAD-DPM	87.41	[12.36]	0.90	[0.52, 0.98]	72.07	[9.48]	0.92	[0.54, 1.00]
MAD-ADA	90.61	[10.28]	0.96	[0.77, 1.00]	73.45	[7.69]	0.96	[0.64, 1.00]

NOTE: The results are obtained by averaging 50 simulated datasets. Bold values represent the lowest MSE among the proposed methods and the lowest MSE among the competitors, for each scenario. Standard error and interquartile range are reported in brackets for MSE and median coverage, respectively.

Table 2. Out-of-sample AUC evaluated on 10^4 new observations and median frequentist coverage for the 95% credible intervals for $\mathbb{E}(Y | \mathbf{x})$ for each of the 1024 possible values of the covariate vector $\mathbf{x}$.

Classification	$n = 150$				$n = 300$			
	AUC		Median coverage		AUC		Median coverage	
GLM	0.796	[0.014]	0.18	[0.06, 0.46]	0.809	[0.007]	0.00	[0.00, 0.20]
BART	0.863	[0.026]	0.18	[0.00, 0.71]	0.932	[0.009]	0.22	[0.00, 0.80]
RF	0.882	[0.025]	0.18	[0.10, 0.38]	0.913	[0.015]	0.70	[0.60, 0.84]
DP	0.644	[0.011]	0.00	[0.00, 0.00]	0.724	[0.012]	0.00	[0.00, 0.02]
MAD-1	0.873	[0.014]	0.44	[0.08, 0.73]	0.899	[0.008]	0.34	[0.02, 0.68]
MAD-2/3	0.869	[0.015]	0.86	[0.72, 0.96]	0.899	[0.009]	0.98	[0.92, 1.00]
MAD-DPM	0.872	[0.014]	0.72	[0.52, 0.89]	0.901	[0.008]	0.74	[0.48, 0.90]
MAD-ADA	0.874	[0.014]	0.82	[0.62, 0.94]	0.900	[0.008]	0.90	[0.72, 0.98]

NOTE: The results are obtained by averaging 50 simulated datasets. Bold values represent the highest AUC among the proposed methods and the highest AUC among the competitors, for each scenario. Standard error and interquartile range are reported in brackets for AUC and median coverage, respectively.

Random forests produce overly conservative credible intervals, with coverage levels substantially exceeding 0.95.

4.2. Classification with Binary Covariates

We also conducted a simulation study focused on binary classification from 10 binary covariates. We generated 50 simulated datasets for each sample size $n = \{150, 300\}$. We sample $y_i | \mathbf{x}_i$ from $\text{Bin}\{1, \text{logit}^{-1}(\eta_i)\}$, with

$$\eta_i = -3 + 2x_{i,1} - 4x_{i,2} + 3x_{i,3} - 3x_{i,4} - 3x_{i,5}x_{i,6} \\ + \sqrt{|2x_{i,7} - 3x_{i,8}|} + (3x_{i,9} - 2x_{i,10})^2,$$

$i = 1, \dots, n$, and draw $x_{i,j} \sim \text{Bernoulli}(\tau_j)$, $j = 1, \dots, 10$, with $\boldsymbol{\tau} = (0.45, 0.65, 0.7, 0.4, 0.4,$

$0.6, 0.7, 0.3, 0.55, 0.55)$ independently. We considered the variants of MAD sequences described in Section 4.1 for count data regression. The hyperparameters for the binary base kernels for the covariates are set to the default value of 0.25, while the response hyperparameter is selected by maximizing the prequential log-likelihood. We compare with DP, logistic GLM, BART, and random forest. We evaluated out-of-sample predictive accuracy using the ROC curve computed on 10^4 new observations, and compared the corresponding area under the curve (AUC). Uncertainty quantification performance is evaluated as in Section 4.1. The results are reported in Table 2.

In terms of predictive accuracy, MAD sequences outperform both the DP and logistic GLM models, and are competitive with BART and random forests, particularly when $n = 150$. For $n = 300$, BART and random forests produce slightly more accurate

point predictions; however, the credible intervals produced by MAD sequences are significantly better calibrated in terms of frequentist coverage. In particular, although we do not claim optimal calibration guarantees, MAD sequences with weights $w_n = (\alpha + n)^{-2/3}$, as well as those using adaptive weights, produce 95% credible intervals with empirical coverage close to the nominal level, especially when $n = 300$. In contrast, while providing good point predictions, the standard choice $w_n = (\alpha + n)^{-1}$ results in poorly calibrated credible intervals.

5. Application: Corvids Abundance in Finland

We analyze data from an ecological monitoring study of birds in Finland (Lindström et al. 2015), focusing on the occurrence rates of corvids in year 2009. Specifically, four species of the survey belong to the corvid family (*Corvidae*): carrion crow (*Corvus corone*), Eurasian magpie (*Pica pica*), common raven (*Corvus corax*), and Eurasian jay (*Garrulus glandarius*). Corvids are non-migratory birds that occur throughout Finland. Monitoring their abundance is particularly important, as they can respond to habitat and climate changes more rapidly than most other species (Jokimäki, Kaisanlahti-Jokimäki, and Suhonen 2022). Moreover, their presence influences ecological balance, given their efficiency as predators and their ability to live in close associations with humans (Madden, Arroyo, and Amar 2015). The study includes observations from $n = 76$ different sites. For each observation, the dataset includes environmental covariates that include habitat type (broad leaf forest, coniferous forest, open habitat, urban habitat, or wetlands) and temperature in April-May (classified as cold, mild or warm if the registered

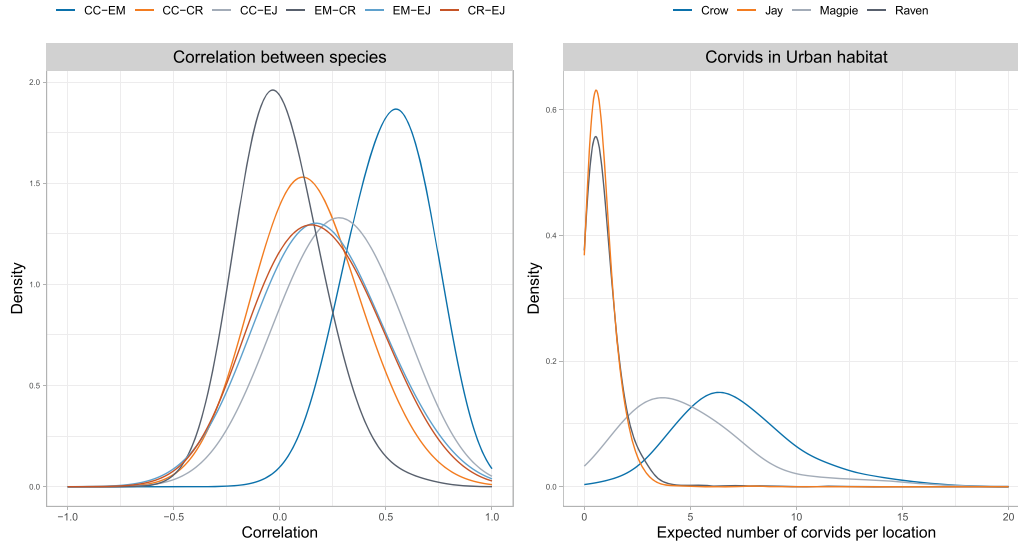


Figure 5. Left: posterior distribution for the correlation across species, with CC, EM, CR, and EJ denoting common crow, Eurasian magpie, common raven, and Eurasian jay, respectively. Right: posterior distribution of the expected number of corvids observed at each urban habitat location.

average temperature is below 5°C , between 5° and 7.5°C , or above 7.5°C , respectively).

Modeling bird species abundance data presents several challenges: count data are often zero inflated and overdispersed relative to the Poisson distribution, and the dependence structure between species is unknown a priori. MAD sequences are particularly well suited for modeling such data, as their flexibility allows us to effectively address zero inflation and overdispersion while capturing across species dependence through the multivariate MH kernels described in Section 3. We compare MAD sequences with three widely used methods in ecological modeling: hierarchical modeling of species communities (HMSC; Ovaskainen et al. 2016; Tikhonov et al. 2017; Ovaskainen and Abrego 2020), generalized joint attribute models (GJAMS; Clark et al. 2017), and generalized linear latent variable models (GLLVMs; Skrondal and Rabe-Hesketh 2004). These models capture species dependence by incorporating latent variables in the linear predictor, and HMSC additionally includes site-specific random intercepts. To account for overdispersion and zero inflation, we consider negative binomial (GLLVM-NB) and zero-inflated negative binomial (GLLVM-ZINB) distributions for the response in GLLVMs.

For the MAD sequence, we choose the adaptive weights $w_n = (\alpha + n)^{-\lambda_n}$, with λ_n defined in (12). We use a weakly informative uniform base measure setting $\alpha = 1$. As default choice we set $\lambda = 3/4$ and $N_* = 500$, as it also proved to be effective in the simulation study of Section 4.1. Then, we select hyperparameters by minimizing the prequential log-likelihood, obtaining $\delta = (0.45, 0.41)$ for temperature and habitat, respectively, and $\sigma = (3.81, 1.53, 1.12, 1.06)$ for the response. For permutation invariance, we average the MAD predictive and the corresponding prequential log-likelihood over 10 permutations. Computing the MAD predictive distribution averaged over permutations takes 25.68 sec on a 3.2 GHz 8-Core Apple M1 CPU. Uncertainty quantification is carried out using a predictive resampling algorithm with $N = n + 5000$ and $B = 1000$. To compare methods, we evaluate out-of-sample predictive accuracy using

predictive mean squared error (MSE) computed with respect to the data obtained at the same locations for year 2010. As a point estimate for MAD sequences we consider the population mean computed at time n . The MAD sequence had the best predictive performance ($\text{MSE} = 3.40$), followed by GJAM (3.78), GLLVM-ZINB (4.69), HMSC (4.70), GLLVM-NB (4.72). Moreover, as shown in supplementary material S3, the MAD sequence performs well in capturing the dependence structure between species. In contrast, popular parametric latent variable models built from elaborations of hierarchical GLMs struggle to recover the empirical dependence structure.

The results obtained using MAD sequences provide valuable insights. The right panel of Figure 5 presents the posterior distribution of the population mean for each species in each urban location, obtained as discussed in Section 2.3. The figure suggests that there are more crows and magpies (95% credible intervals $[3.15, 11.28]$ and $[0.70, 8.81]$, respectively) in urban habitats compared to ravens and jays (95% c.i. $[0.06, 1.61]$ and $[0.11, 1.57]$, respectively). This aligns with Matsyura, Jankowski, and Zimaroyeva (2016) and Jokimäki, Kaisanlahti-Jokimäki, and Suhonen (2022), which indicate that urban settlements serve as stable wintering environments for magpies and crows. As shown in the left panel of Figure 5, there is a positive dependence between these two species (posterior mean for correlation is 0.45, with 95% c.i. $[0.24, 0.69]$), probably due to shared habitat preferences. A similar result was observed by Dupak and Telizhenko (2023) for hooded crows (a subspecies of carrion crows) and magpies in Ukraine. Moreover, urban environments are also the preferred habitat for magpies (as reported in supplementary material S3), probably because urbanization buffers seasonal variations in climate conditions and food availability (Benmazouz et al. 2021), facilitating magpies adaptation. However, within the urban environment, Eurasian magpies avoid densely populated areas (Jokimäki et al. 2017). The differences in the estimated distributions at different temperatures represented in Figure 6 suggest that carrion crows prefer areas with higher temperature, which correspond to the Southern Finland region

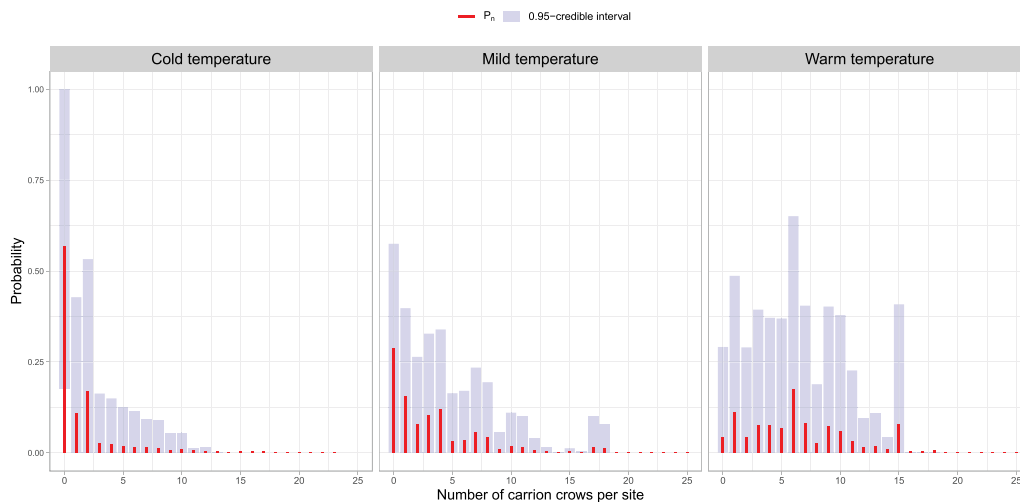


Figure 6. MAD predictive distribution of carrion crows across temperatures (red) with corresponding 0.95 credible interval (blue).

(see supplementary material S3). Finally, consistent with Madge and Burn (1994), our results indicate that the distributions of common ravens and Eurasian jays are homogeneous across different habitats and temperature conditions.

6. Discussion

Our proposed MAD sequences provide a flexible and effective approach for nonparametric modeling of discrete data distributions. We have shown that these sequences implicitly define a prior over the data-generating mechanism which, unlike the DP, can induce positive correlations between nearby values in the support, making MAD sequences appealing in practical applications.

As discussed in Section 2.4, the weight sequence $(w_n)_{n \geq 1}$ plays a dual role, governing both the posterior variability induced by the MAD prior and the learning rate of the associated predictive rule. Although the adaptive weights perform well empirically and represent a valid choice for practitioners, theoretical guarantees ensuring well-calibrated frequentist coverage of posterior credible intervals remain to be established for any available choice of weights. While this issue can be addressed in parametric settings (Fong and Holmes 2021; Fong and Yiu 2024a), it remains an open challenge in the context of Bayesian nonparametric predictive inference, representing an important area for future research (Fortini and Petrone 2025).

While MAD sequences are designed for discrete distributions, they could, in principle, also be employed to model continuous data by using a continuous base kernel. Although the resulting MH kernel would be a mixture of a continuous base kernel and a point mass at the observed data point—which is undesirable when modeling continuous variables—they might be used in a regression setting with continuous covariates and discrete response, where one could employ a continuous MH kernel for modeling the covariates to guarantee dimension-ordering invariance while modeling $p(y | \mathbf{x})$ using a discrete MH kernel.

Although MAD sequences have proven effective for modeling multivariate count data, their performance is likely to degrade in high-dimensional cases due to the curse of dimensional-

ity. A promising direction for future work is to incorporate MAD sequences into more structured models—for example, by using them to define priors over latent parameters. See Airolidi et al. (2014) for a precursor to these ideas. In particular, CID sequences could be employed to model low-dimensional latent variables underlying high-dimensional observations. Interesting next steps include establishing theoretical guarantees for these approaches and ensuring their computational scalability.

Supplementary Materials

The supplementary material contains the proofs of all the Theorems and Propositions, a simulation comparing MAD sequences with the copula updating approach, and additional figures. Code and datasets to replicate our analyses is available online in the repository https://github.com/davideagno/MAD_sequences.

Disclosure Statement

The authors report there are no competing interests to declare.

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ORCID

Davide Agnoletto  <https://orcid.org/0009-0002-3067-0745>
Tommaso Rigon  <https://orcid.org/0000-0002-9224-543X>

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